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Tasks:

A. Discuss Briefly:

1. What is an algorithm, and how is it related to the tasks you perform in your daily life?

Based on the Youtube video I have watched, an algorithm is a set of clear, step-by-step instructions used to complete a task or solve a problem. The video explains that algorithms are not only used by computers but can also be seen in our daily activities, such as following the steps for brushing our teeth, cooking food, or going to school. Just like for example, when I prepare for school, I follow step-by-step processes such as waking up, taking a bath, eating breakfast, brushing my teeth, preparing my things, and going to school. These steps are done in an order to reach my goal to go to school, just like how an algorithm helps a computer complete a task.

2. Why is it important for an IT student to learn algorithms?

It is very important for an IT student like me to learn about algorithms because they are the building blocks of programming and serve as a blueprint for creating a program or system. An algorithm is often the very first step where I can plan a clear, step-by-step procedure for solving a problem before making a system, program, or application. Just like a blueprint guides the construction of a house, an algorithm guides me in building a program and helps me organize its logic before writing the actual code. Learning algorithms also helps me develop better problem-solving and logical-thinking skills, which are important in becoming a good IT professional

B. Convert each JavaScript code into a step-by-step algorithm using simple English or pseudocode.

Guide: Describe what the program does from the beginning until the loop stops. You may use numbered steps.

```
for (let i = 1; i < 10; i *= 2) {
  console.log(i);
}
```

Step-by-Step Process

1. Start with i = 1.
2. Check if i is less than 10.
3. If yes, display the value of i.
4. Multiply the value of i by 2.
5. Check again if i is less than 10.
6. Repeat the steps until i reaches 10.
7. Stop the loop.

Output (1, 2, 4, 8)

```
for (let i = 0; i < 10; i++) {
  console.log(i);
}
```

Step-by-Step Process

1. Start with i = 0.
2. Check if i is less than 10.
3. If yes, display the value of i.
4. Add 1 (increment) to the value of i.
5. Check again if i is less than 10.
6. Repeat the steps until i reaches 10.
7. Close the loop.

Output (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

C. Run both JavaScript codes. Based on the outputs and execution of the two codes, answer the following:

1. What did you observe about the sequence of numbers produced by the first code?

Based on what I observed, the numbers start at 1 and double each time, giving 1, 2, 4, and 8.

(|1<10? 1*2=| 2<10? 2*2=4 | 4<10? 4*2=8|)

The loop stops after 8 because the next number, 16, is already greater than 10.

2. What did you observe about the sequence of numbers produced by the second code?

I observed that the numbers start at 0 and increase by 1 each time. The sequence is 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.

(|0<10? 0+1=1,| 1<10? 1+1=2, |2<10? 2+1=3|3<10? 3+1=4| 4<10? 4+1=5|5<10? 5+1=6|6<10? 6+1=7|7<10? 7+1=8|8<10? 8+1=9|)

The loop stops when i reached 10.

3. What is the main difference between the way the two loops progress?

The main difference is that the first loop multiplies the number by 2 each time, while the second loop adds 1 to the number each time. The first loop goes 1, 2, 4, 8, while the second loop goes 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

4. How does the way an algorithm performs its steps affect its output?

The way an algorithm does its steps affects its output as each step helps decide the final result. If the steps or actions are changed, the output may also change. Adding 1 to the value gives 0, 1, 2, 3, while multiplying the value of i by 2 gives 1, 2, 4, 8 which are different.